Votes Without Neighborhoods: Housing Prices and the Geography of Conservative Voting in Seoul

Abstract. Neighborhood election returns can combine residents' votes with ballots cast by people who live elsewhere. We examine how this reporting problem affects the association between housing prices and conservative voting in six Seoul elections, 2012–2026. Among election-day voters, the positive within-district gradient extends beyond Gangnam, Seocho and Songpa in 2012 and the four sampled elections from 2022. To assess the relationship across voting modes, we bound a resident-based, roll-denominator vote share over allocations consistent with returns, electoral rolls and statutory voting zones. Returns alone leave the sign unresolved in every sampled election with nationwide early voting. Under the main movement and composition restrictions, the gradient is positive in 2026, narrowly positive in 2024, and unresolved in 2022 and 2025. The 2026 result extends beyond the three districts but is sensitive to out-of-zone composition and the highest-price neighborhoods. Holding votes fixed, finer statutory zones can establish the sign under weaker restrictions. Reporting out-of-zone ballots by registered neighborhood would guarantee reductions of 27–57% in the remaining range across the four recent elections under the main restrictions. These results show which housing-related electoral patterns survive uncertain residential attribution and how reporting changes could make neighborhood comparisons more informative.
Keywords: electoral geography; housing politics; Gangnam; early voting; election administration; South Korea

1. Introduction

Housing prices and conservative voting are closely associated in Seoul, but the geographic reach of that relationship is less clear. Gangnam, Seocho and Songpa occupy a distinctive position in the city's housing market and political geography (Son 2008, 2010; Bae and Joo 2020). A citywide association could reflect their contrast with the rest of Seoul, differences among neighborhoods within districts, or both. Establishing a broader pattern also requires identifying whose votes it represents. Early voters differ politically from election-day voters, and their ballots are not always reported where they live (Lee and Kwon 2019; Ka 2021; Han 2024).

The June 2026 mayoral election illustrates the problem. A polling firm's research note reported a correlation of 0.84 between apartment prices and Oh Se-hoon's vote share across 424 administrative neighborhoods (*dong*), and 0.71 within districts (*gu*) (Lee 2026). The fine geographic detail suggests a precise account of local voting. Yet a neighborhood's published total combines votes cast by some of its residents with votes cast there by residents of other neighborhoods, and omits ballots reported only at a larger scale. Since nationwide early voting began in 2014, a substantial share of Seoul's ballots has been affected. In recent elections, 22–35% were cast early inside the voter's statutory zone and counted at the station's neighborhood. Another 12–15% were out-of-zone, at-home or overseas ballots reported only for a district

or constituency. Only election-day ballots are attributed exactly to residential neighborhoods. Conservative support among in-zone early voters averaged 4 percentage points below the election-day share in the same district in 2017, and 10, 12, 28 and 26 points below it in 2022, 2024, 2025 and 2026 (Online Appendix Table C8). Korean and comparative research documents political selection and participation differences under early and convenience voting (Ka 2021; Gronke et al. 2008; Kaplan and Yuan 2020). Assuming that these voters share election-day voters' neighborhood pattern can therefore change the substantive conclusion.

We examine six Seoul elections from 2012 to 2026. We distinguish associations across districts from those among neighborhoods within a district, then repeat the comparisons without Gangnam, Seocho and Songpa. For residents across all voting modes, we calculate the smallest and largest price–vote slopes consistent with published candidate counts, electoral rolls and statutory voting zones. These bounds show what the returns establish and how further restrictions on early-voter movement and composition change the answer.

The price–vote association across districts is positive in all six elections. Within districts, the election-day gradient is positive in 2012 and the four sampled elections from 2022, including outside the three southeastern districts, and nearly flat in 2017. Extending that result across voting modes produces a different picture. Returns alone do not establish the sign after nationwide early voting began. Under our main assumptions, the slope is positive in 2026, narrowly positive and sensitive to specification in 2024, and unresolved in 2022 and 2025. Outside the three districts in 2026, a doubling of neighborhood prices corresponds to a fitted conservative roll-based share 2.0 to 9.6 points higher. This extension is more sensitive to out-of-zone composition and high-price neighborhoods than the whole-city result.

Reporting rules help explain what can be learned from these returns. Applying the finer statutory zones to the same 2026 votes narrows the main range by about a quarter and can establish its sign under weaker restrictions. Reporting out-of-zone votes by residential neighborhood would guarantee a reduction of 27–57% in the remaining range across the four recent elections under the main assumptions. These are information gains from fixed votes, so they separate what administration reveals from changes in political behavior.

The paper contributes to housing politics by establishing where the observed association extends beyond Seoul's southeastern enclave and which claims about the wider electorate require assumptions about early voters. It also addresses a measurement problem distinct from changing the size or boundaries of spatial units (Lee et al. 2025): ballots can be attributed to the wrong neighborhood even when boundaries remain fixed. Related problems arise for German postal ballots, Australian declaration votes and American precinct datasets that allocate absentee votes (Online Appendix B.5; Amos et al. 2017; Baltz et al. 2022). Explicit allocation bounds allow those residential claims to be assessed without selecting a single unobserved distribution of votes.

2. Housing and conservative voting in Seoul

Housing is a central axis of Korean political conflict. Son's (2008, 2010) work on Seoul and the capital region found that homeownership, apartment residence and housing values tracked conservative voting across neighborhoods, a "real-estate class society". Survey research relates homeownership to vote choice (Lee and Kwon 2019; Shin 2022) and subjective class position to conservative support (Han 2026), and studies of administrative units relate housing values to conservative support across municipalities, dis-

tricts and constituencies (Oh and Park 2013; Han 2024; Song and Kang 2025). In the transactions we use, the median neighborhood's price per square meter roughly doubled between the twelve months before the 2017 and the 2022 presidential elections, and the conservative opposition won the 2021 Seoul mayoral by-election and the 2022 presidential election. Comparative work links housing wealth and local housing markets to political preferences and populist support (Ansell 2014, 2019; Larsen et al. 2019; Adler and Ansell 2020; Ansell et al. 2022), and in the United States buying a home increases turnout in local elections (Hall and Yoder 2022).

Existing studies address different spatial scales. Han (2024) relates conservative support to average apartment prices across districts, treating housing wealth as a regional characteristic. Bae and Joo (2020) explain Gangnam's conservatism through the construction of place identity: residents' efforts to preserve the area's exclusivity and transmit it to their children. Their account raises the question of how much a city-wide price gradient depends on this distinctive area. Dong-level studies examine finer spatial differences, relating price levels or price changes to electoral outcomes (Kim and Kang 2022; Yun and Jun 2024; Lee 2026). Such studies address related but distinct quantities; an association involving price changes does not directly test the price-level gradient examined here. Substantial variation within districts makes the latter question relevant: in our 2026 transaction data, median apartment prices per square meter differ by a median of 24% between adjacent neighborhoods in the same district, and 39% of the variance of log neighborhood prices lies within districts.

Owners' economic interests (Fischel 2001; Marble and Nall 2021), residential sorting (Rodden 2019) and place identity offer different explanations for the observed relationship. Neighborhood prices may reflect the households that select into an area, their shared policy interests, or the identity attached to living there. These mechanisms can also operate outside wealthy enclaves and vary with local conditions (Hankinson 2018). Finding a price gradient beyond Gangnam would therefore extend the geography to be explained, without choosing among these mechanisms.

The comparisons answer progressively narrower empirical questions. District-level estimates describe differences across Seoul; district fixed effects ask whether more expensive neighborhoods also vote more conservatively than other neighborhoods in the same district. Excluding Gangnam, Seocho and Songpa tests how much that relation depends on the southeastern enclave. Finally, comparing election-day voters with residents across all voting modes tests whether the pattern extends to politically different early voters. Income, age and residential selection remain possible explanations of the neighborhood association. The analysis identifies its geographic and electoral reach, not a causal effect of housing wealth.

The primary estimand is the *gradient*: the slope of the conservative share on log apartment price, with a separate intercept for each district and equal weight for each priced neighborhood. It measures the price–vote relation within districts. A slope of 0.1 corresponds to a fitted share about 6.9 percentage points higher when prices double. Neighborhoods far from their district's mean log price contribute more to the slope, so Section 4.5 examines the influence of the most expensive neighborhoods. Secondary *contrasts* compare higher- and lower-price groups within each district, using the median split and top versus bottom third, then average those differences with weights proportional to each district's number of priced neighborhoods. The slope and contrasts summarize different features of the same association and can yield different sign conclusions.

3. Electoral reporting, data and identification

3.1 Electoral reporting rules

Korea reports election results by voting district and therefore by administrative neighborhood, a unit with about 19,500 electors in Seoul. Box 1 and Figure 1 summarize where each kind of ballot is counted. Election-day ballots are counted in the voter's own voting district. Since the April 2013 by-elections and nationwide since 2014, any voter can also vote early at any station in the country. In-zone early voters cast a ballot within their designated zone, placing it directly in the station's ballot box; it is then counted for the station's neighborhood. The statutory zone coincides with the district only in some elections; the terms in-zone and out-of-zone refer to that election-specific boundary. Out-of-zone early voters seal their ballot in an envelope that is mailed to their district commission and counted only for the district, as are at-home and overseas ballots. Before 2014, voters who could not vote on election day registered as absentees and their ballots were likewise reported only for the district; only 3.4% of Seoul ballots were pooled in 2012.

Box 1. Where Korean ballots are reported

Legal sources: Public Official Election Act (Act; Korea n.d.) and Regulations on Public Official Elections (Regulations; National Election Commission n.d.), in the versions in force at each election.

Ballot	Rule	Where reported
Election day	Counted and published by voting district (Act, Art. 178)	Voting district, hence neighborhood
In-zone early	No envelope inside the zone (Act, Art. 158(5); zone: Regulations, Arts. 86(3), 133(3)); each station's box is counted separately	Station's neighborhood
Out-of-zone early	Envelope mailed to the voter's district commission (Act, Arts. 158, 176)	District or constituency
At-home, shipboard, overseas	Registered in advance; mailed to the district commission	District or constituency
Election-day roll	Printed after early voting, early voters marked (Regulations, Art. 16-2)	Residents removed from the roll

The early-voting zone is decisive for our question, because a voter can be misattributed to another neighborhood only inside it. In presidential elections the zone is the whole district, about 17 neighborhoods; in National Assembly elections it is the constituency where a district holds two or more (Regulations, Art. 86(3)). In the 2026 local elections it was the district-council constituency (Regulations, Art. 133(3); National Election Commission 2022a): 155 constituencies in Seoul, with 2.8 neighborhoods on average. The council returns are consistent with this restriction: at all 320 early-voting stations in contested constituencies, the number of early voters in the council contest equals the number in the mayoral contest.

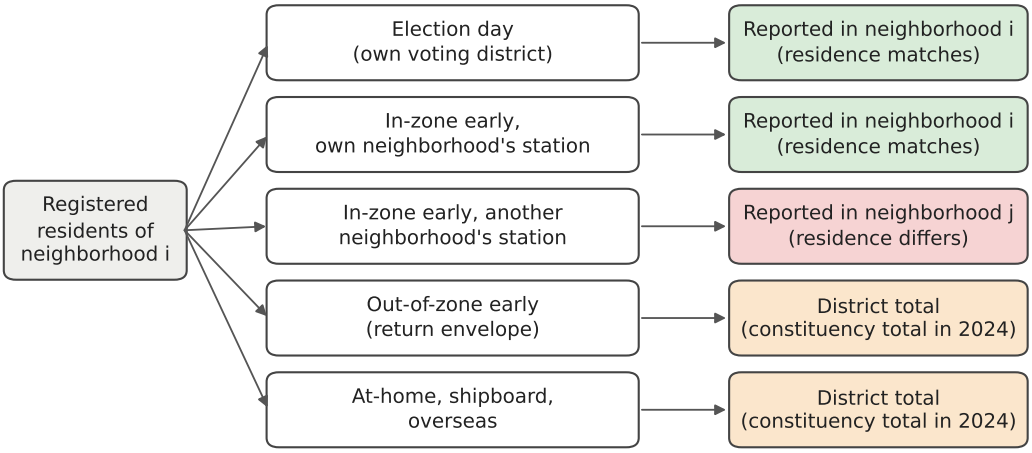


Figure 1: Where a Seoul resident's ballot is reported, by voting mode, since 2014. The zone is the district; in National Assembly elections, the constituency where a district holds two or more; in local elections, the district-council constituency.

Two features of the published data make inference possible. The election-day roll is printed after early voting with early voters marked, so the difference between each neighborhood's electorate and its election-day roll counts the residents removed from that roll: those who voted early or were registered to vote at home or abroad, some of whom cast no counted ballot. And pooled-ballot totals are published by candidate for each counting jurisdiction. With each station's count, these numbers constrain how the unattributed ballots can have been distributed.

3.2 Data

We study the presidential elections of 2012, 2017, 2022 and 2025, the 2024 National Assembly election and the 2026 election of the mayor of Seoul (Table 1). The four presidential contests permit same-office comparisons. The legislative and mayoral contests provide different early-voting zones: parliamentary constituencies in 2024 and smaller district-council constituencies in 2026. The dataset covers these six contests; the cross-election comparisons do not estimate a continuous trend (Online Appendix B.1).

Returns and electorates are the National Election Commission's official counting-unit files and electorate records (National Election Commission 2026). The 2026 zones come from its official table of council constituencies and neighborhoods. We measure support for the main conservative candidate: Park Geun-hye (2012), Hong Joon-pyo (2017), Yoon Suk Yeol (2022), the People Power Party's constituency candidates (2024), Kim Moon-soo (2025) and Oh Se-hoon (2026). In 2017 the conservative vote was split. Combining Hong with Yoo Seung-min also leaves the within-district sign unresolved under returns alone and minimal movement (Online Appendix Table C24).

Neighborhood prices are medians of apartment sale prices per square meter during the twelve complete calendar months before each election month, from Ministry of Land, Infrastructure and Transport (2026) transactions geocoded to neighborhoods. They measure traded apartments, not residents' ownership or wealth. The movement scenarios use Seoul Living Mobility data, which estimate trips from mobile-phone signals and are available from 2022 (Seoul Metropolitan Government and KT 2026). Table 1 gives price coverage and the shares of votes requiring residential allocation; Online Appendix B documents the sources and construction.

Table 1. Elections, reporting units and ballot coverage

Election	Pooled ballots reported for	In-zone early zone	Dongs	Dongs with price	Pooled share	In-zone early share	Out-of-zone share of early ballots	Stations with forced movement
2012 pres.	25 gu	–	423	414	3.4%	–	–	0
2017 pres.	25 gu	25 gu	424	421	12.8%	21.7%	34.6%	5
2022 pres.	25 gu	25 gu	425	420	13.5%	35.3%	26.1%	20
2024 leg.	48 constituencies	48 constituencies	425	418	12.9%	34.8%	26.1%	20
2025 pres.	25 gu	25 gu	426	420	14.7%	29.2%	31.7%	13
2026 mayor	25 gu	155 council constituencies	427	423	12.1%	25.6%	31.8%	8

Notes. Shares are of all ballots cast in Seoul. Pooled ballots: out-of-zone early, at-home, shipboard and overseas. "Out-of-zone share of early ballots" is P/(P+K) over Seoul. Forced movement: stations whose in-zone early ballots exceed their dong's removed residents ($r_j > 0$). 2012 had no in-zone early voting.

3.3 Identification

An illustrative district.

Consider a district with two neighborhoods, *a* and *b*, each with 1,000 residents who cast a ballot. Of these, 600 in each neighborhood voted on election day, and candidate X received 330 ballots in *a* and 270 in *b*. Each neighborhood's early station counted 300 ballots, 150 for X at *a*'s station and 120 at *b*'s. The district reports 200 out-of-zone early ballots, 120 of them for X. Suppose 300 residents of each neighborhood voted early in the zone and 100 outside it.

Two things are unknown. *Movement*: if 30 residents of *a* voted at *b*'s station and 30 residents of *b* at *a*'s, the station counts are unchanged, but *a*'s residents may have cast anywhere from 30 more to 30 fewer X ballots than its station shows, a change of 3 percentage points in *a*'s share. *Pooling*: the 120 pooled X ballots may be split so that *a* receives anywhere from 20 to 100 of them, a further 8-point range. The district's share of X is the same in every case; how it divides between *a* and *b* is not. A conventional map treats station counts as residential counts and drops pooled ballots, leaving both sources of uncertainty unrepresented.

Feasible allocations.

An allocation must reproduce the election-day and station counts, the residents removed from each roll, and pooled candidate totals, while respecting statutory zones. We minimize and maximize the within-district slope over these allocations. A positive lower bound means that every compatible allocation yields a positive slope; a range spanning zero leaves the sign unresolved. Linear programming gives sharp bounds for continuous ballot counts and conservative outer bounds for integer counts (Manski 2003; Tamer 2010). Under the main assumptions, integer witnesses for the four elections from 2022 place endpoints within 0.0008 of the continuous bounds in both geographic scopes (Online Appendix A.4).

Conventional ecological inference recovers unobserved cross-classifications from observed area totals (Duncan and Davis 1953; King 1997; Wakefield 2004; Cho and Manski 2008; Imai et al. 2008; Greiner and Quinn 2009; Jiang et al. 2020). Here the residential totals themselves are unobserved because reporting intervenes between voting and the record (Knox et al. 2020; Cross and Manski 2002; Molinari and Peski 2006; Ridder and Moffitt 2007). Like recent bounds using monotonicity restrictions (Elzayn et al. 2026), our analysis makes each assumption's identifying role explicit, but addresses attribution before individual-level inference.

3.4 Assumptions and outcomes

The published counts permit extreme allocations, including ones in which neighboring areas' residents account for almost entirely different candidates' out-of-zone ballots. To assess how much the conclusions depend on such possibilities, we impose three restrictions and vary their strength (Manski and Pepper 2018; Rambachan and Roth 2023).

1. *Cross-neighborhood movement.* At the lower extreme, early voters moved only as much as the roll forces: when a station counted more ballots than its neighborhood had residents removed from the roll, the difference must have come from neighbors. We call this the *minimal-movement benchmark*; it serves as a lower-movement reference rather than an estimate of actual behavior. For a *mobility-calibrated scenario*, we use mobile-phone trip data (Seoul Living Mobility) to calibrate an upper bound on cross-neighborhood movement: beyond a baseline determined by the rolls and zone totals (Online Appendix A.1), the early voters who voted at another neighborhood's station in their zone may number at most the zone's out-of-zone early voters multiplied by the ratio of residents' daytime trips to other neighborhoods of the zone to their trips leaving it. Trips are not votes, so the data set the scale of a cap rather than measure movement; studies that infer turnout from phone visits to polling stations use such data differently (Harada et al. 2025). We vary the cap by a factor λ and also report unrestricted movement.
2. *Out-of-zone composition.* We allow the candidate shares of a neighborhood's out-of-zone ballots to differ from those of its early station by at most δ, ten points in the main specification. A version that first shifts the station shares by the district's observed difference between out-of-zone and in-zone early voters yields nearly identical results.
3. *Residents' in-zone early-vote composition.* We benchmark each neighborhood's early vote by its own election-day vote, shifted by the district's observed gap between early and election-day voters. We then allow residents' early ballots, wherever in the zone they were cast, to depart from this benchmark as much as their station's ballots do, plus an additional tolerance s (two percentage points in the main specification). For example, if a station's share differs from the benchmark by three points, $s = 0.02$ permits its residents' in-zone early share to differ by up to five points. This restriction carries much of the election-day pattern over to early voters: with small s, residents' early ballots inherit their neighborhood's election-day vote, shifted by a district-wide gap. Its role is therefore to ask how far the early-vote gradient could differ from the election-day gradient, which is point-identified (Section 4.2), before a conclusion about all residents changes.

We call the combination of the mobility-calibrated scenario, $\delta = 0.1$ and $s = 0.02$ our *main assumptions*. They are reference points, not validated error bounds. Stations' own ballots depart from the benchmark by a median of about one point, and in the five elections with early voting the conservative share among a district's out-of-zone early voters differs from that among its in-zone early voters by at most 7.7 points (Online Appendix Table C21), so the 10-point tolerance exceeds every district-level gap observed. What δ bounds, however, is the gap for each neighborhood's residents, which is not observed and may vary within a district more than district totals show. Moreover, the population classified as out-of-zone differs across election types. We therefore treat as evidence the range of settings under which a conclusion holds: each assumption is indexed by one parameter, so we report how far it can be relaxed before a conclusion changes and what the allocation that first reverses it looks like (Masten and Poirier 2020; Table 3, Panel

B; Section 4.5). Box 2 states the accounting identities and the outcome; Online Appendix A gives the assumptions in the same notation.

Table 2 separates the contribution of each restriction: counts and rolls alone, movement and out-of-zone composition, then residents' in-zone composition. The last restriction links an unobserved group to the observed election-day pattern. Sensitivity thresholds show how far that link can be relaxed; they neither validate it nor assign probabilities to the assumed tolerances.

Box 2. The accounting identities and the outcome

For each counting jurisdiction (a district, or a constituency in 2024), let A_{ic} be the election-day ballots for candidate c in neighborhood i, K_{jc} the in-zone early ballots counted at the station of neighborhood j, E_i the registered electorate, D_i the electorate remaining on the election-day roll, and P_c and Q_c the jurisdiction totals of out-of-zone early ballots and of at-home and overseas ballots. The difference $E_i - D_i$ counts residents removed from the election-day roll. The unknowns are F_{ijc}, the early ballots of residents of i cast at station j; P_{ic} and Q_{ic}, residents' pooled ballots; and U_i, removed residents who cast no counted ballot. Every allocation must satisfy

$$\textstyle\sum_{j,c} F_{ijc} + \sum_c P_{ic} + \sum_c Q_{ic} + U_i = E_i - D_i, \qquad \sum_i F_{ijc} = K_{jc}, \qquad \sum_i P_{ic} = P_c, \qquad \sum_i Q_{ic} = Q_c,$$

and $F_{ijc} = 0$ when i and j lie in different early-voting zones. Residents' votes for c are $N_{ic} = A_{ic} + \sum_j F_{ijc} + P_{ic} + Q_{ic}$.

The outcome. With A_i the election-day voters of i, let $H_i = A_i + (E_i - D_i)$ and let $I_i = N_{i,\text{inv}}$ be residents' invalid ballots. We analyze the roll-based share $z_{ic} = N_{ic}/H_i$, whereas the usual valid-vote share is $y_{ic} = N_{ic}/(H_i - U_i - I_i)$. Because H_i includes voters casting invalid ballots and removed residents who cast no counted ballot, z is no greater than y and is not the valid-vote share. Its linearity gives exact ranges for continuous allocations and outer bounds for integer ballots. For y we bracket the endpoints. The main-assumption sign conclusions agree, but the 2025 positive slope under minimal movement and a ten-point out-of-zone tolerance does not survive this outcome change (Online Appendix Tables C15 and C18).

For each allocation, the within-district slope is

$$\beta = \frac{\sum_i (x_i - \bar{x}_{g(i)}) z_{ic}}{\sum_i (x_i - \bar{x}_{g(i)})^2},$$

where x_i is log apartment price, $g(i)$ is the district containing neighborhood i, and $\bar{x}_g$ is the mean log price of the priced neighborhoods in that district. The sums run over neighborhoods with price data in the specified sample. Prices and weights stay fixed while the allocation changes z; the two linear programs find the smallest and largest β. Multiplying a slope by $100 \log 2$ expresses the fitted difference for a doubling of prices in percentage points.

The estimands describe registered residents of places, not homeowners, and they are associations, not effects of housing wealth. All results are conditional on the observed returns and price measure. The ranges describe allocation uncertainty for the observed finite population, not confidence intervals: the returns are a census, so this target does not involve sampling voters. This does not eliminate uncertainty from price measurement, address matching or administrative records, nor justify inference to other elections. The election-day slopes are descriptions of the observed neighborhoods; permutation p-values and clustered standard errors are auxiliary diagnostics requiring further assumptions (Abadie et al. 2020; Online Appendix Table C20). Transaction resampling shows sensitivity to the price measure without supplying

confidence intervals with established coverage (Online Appendix Table C13). Inference for estimated bounds is developed elsewhere (Imbens and Manski 2004; Stoye 2009; Fang and Santos 2019; Fang et al. 2023).

4. Results

4.1 The association across districts

At the district scale, ballots can be assigned to the residential unit without neighborhood-level allocation assumptions. The 2024 analysis uses 24 units, merging Jung-gu and Seongdong-gu because one constituency spans both; the other elections use 25 districts. For all of Seoul, higher apartment prices are associated with greater conservative support in every election (Figure 2A; Table 3). A doubling of the geometric mean of a district's neighborhood prices is associated with a conservative share 4 percentage points higher in 2017 and 9 to 14 points higher in the other elections. Excluding Gangnam, Seocho and Songpa reduces the slope to about half to two-thirds of its full value in 2022–2026. In 2012 and 2017 it falls to 0.057 and 0.025, less than half its full value, and model-based standard errors would not distinguish these slopes from zero (Online Appendix Table C14). The earlier district-level associations thus rest largely on the three southeastern districts.

4.2 The association within districts

Table 2 follows the within-district gradient as assumptions are added. Each range contains the slopes consistent with the data and stated assumptions. A range that includes zero leaves the direction unresolved. Table 3 compares the geographic scopes and price-group contrasts; the full contrast results are in Online Appendix Table C11.

Table 2. Within-district price gradients under alternative assumptions

Election	Election day (exact y)	(A) Returns and rolls	(B) Minimal movement	(C) + out-of-zone restriction	(D) Calibrated movement	(E) + residents' composition	(F) Price-dependent benchmark
2012	0.087	[0.008, 0.157]	[0.008, 0.157]	[0.027, 0.140]	–	–	–
2017	0.013	**[−0.428, 0.450]**	**[−0.136, 0.162]**	**[−0.060, 0.085]**	–	–	–
2022	0.097	**[−0.615, 0.704]**	**[−0.100, 0.264]**	[0.014, 0.175]	**[−0.283, 0.448]**	**[−0.026, 0.223]**	**[−0.027, 0.222]**
2024	0.102	**[−0.553, 0.683]**	**[−0.077, 0.254]**	[0.033, 0.164]	**[−0.185, 0.372]**	[0.001, 0.206]	**[−0.003, 0.200]**
2025	0.063	**[−0.468, 0.552]**	**[−0.104, 0.232]**	[0.002, 0.149]	**[−0.283, 0.411]**	**[−0.037, 0.194]**	**[−0.062, 0.183]**
2026	0.086	**[−0.279, 0.439]**	**[−0.046, 0.234]**	[0.072, 0.138]	[0.010, 0.201]	[0.054, 0.156]	[0.052, 0.154]

Notes. Bold ranges include zero; their sign is not identified. Slope of the conservative share among residents (roll-based share z) on log median apartment price per square meter; 0.1 means a neighborhood with prices twice as high has a share about 7 points higher. Within National Assembly constituencies in 2024 the ranges are [−0.729, 0.816] (A), [0.004, 0.166] (C), [−0.269, 0.427] (D) and [−0.035, 0.221] (E). Election day: point-identified valid-vote share among election-day voters (Online Appendix Table C20). Conventional estimates: Table C1. High-minus-low contrasts: Table 3 and Table C11. (A) every allocation consistent with the counts, rolls and statutory zones; (B) minimal movement; (C) adds out-of-zone ballots within 10 points of the station; (D) replaces (B) with mobility-calibrated movement; (E) adds residents' early ballots within their station's departure plus 2 points of the benchmark; (F) price-dependent benchmark. Definitions in Section 3.4 and Online Appendix A.

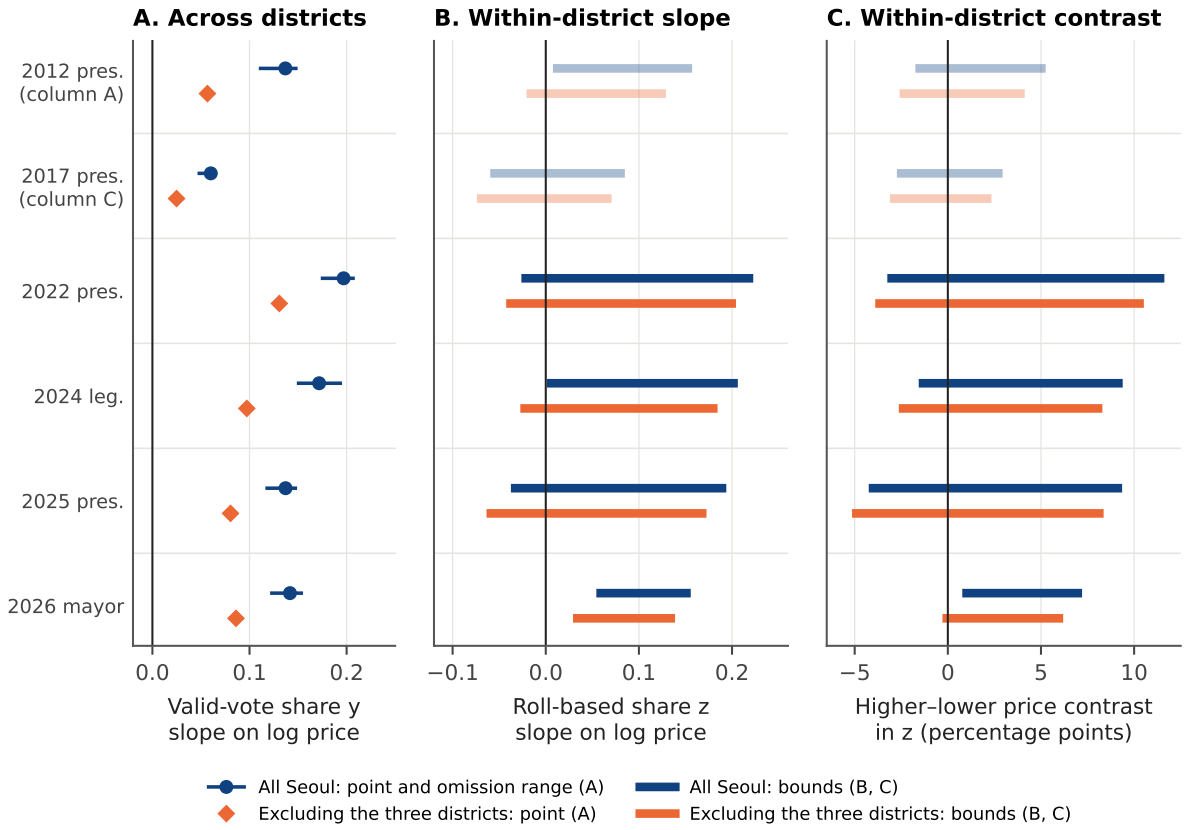


Figure 2: The three spatial claims, for all of Seoul and without Gangnam, Seocho and Songpa. (A) Expensive districts: slope of the conservative valid-vote share on log price across districts; for all of Seoul, the dot is the slope over 25 districts (24 units in 2024) and the line the range of slopes when one district at a time is dropped. (B) Expensive neighborhoods within districts: range of the within-district slope of the roll-based share z consistent with the data under the main assumptions (Table 2, column E). (C) The same for the difference between the more and less expensive halves of each district's neighborhoods, in percentage points. 2012 and 2017, which lack mobility data, use the returns and rolls alone and minimal movement with out-of-zone voters within ten points (columns A and C) and are drawn faint, as they are not comparable with the later elections. Panel A uses valid-vote shares, panels B and C roll-based shares. Ranges are sets of values consistent with the data and assumptions, not confidence intervals.

Election-day voters. Election-day ballots are counted in the voter's own voting district, so the slope of the conservative share among election-day voters needs no allocation assumption (Table 2). It is positive in 2012 and in the four sampled elections from 2022 and close to zero in 2017: 0.097, 0.102, 0.063 and 0.086 in 2022, 2024, 2025 and 2026, and 0.079, 0.078, 0.041 and 0.069 outside Gangnam, Seocho and Songpa (Online Appendix Table C20). The rest of this section asks whether the same holds for all residents, including the third to half who voted early.

The returns alone. The conventional estimates, computed from the published neighborhood rows, are positive in every election. But with the returns and rolls alone, the within-district relation among all residents could be strongly positive or strongly negative in every election since 2014 (column A); in 2022–2026 the ranges are 7 to 14 times as wide as the conventional estimates themselves, and in 2017, when the conventional estimate is close to zero, far wider still. The contrast between higher- and lower-price halves of each district ranges from about −43 to +47 points in 2022 and from −20 to +25 points in 2026, around conventional estimates of 3 to 4 points. Only in 2012, before nationwide early voting, are the returns informative without assumptions: the slope is positive, although the contrast is not established

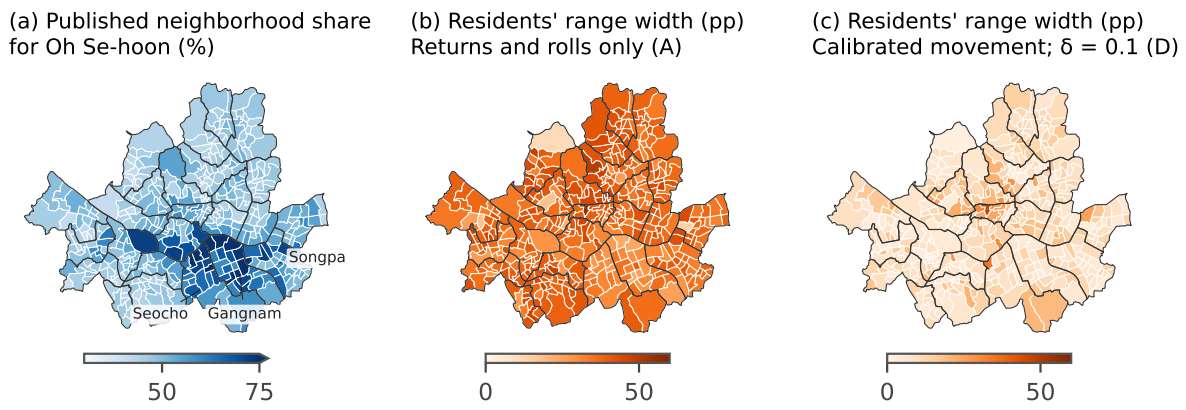


Figure 3: 2026 Seoul mayoral election. (a) Conventional neighborhood share for Oh Se-hoon (published station counts, pooled ballots dropped), in percent. (b) Width, in percentage points, of the range of each neighborhood's residents' share consistent with the returns alone (Table 2, column A). (c) The same under mobility-calibrated movement with out-of-zone voters within ten points of their station (Table 2, column D). Panels (b) and (c) share one scale; pp denotes percentage points. Lines are district boundaries; neighborhood boundaries from vuski (2026).

even then. The returns also say little about individual neighborhoods: with the returns alone, the median neighborhood's conservative share in 2026 is pinned down only within a 39-point range, and the direction of the difference between adjacent neighborhoods is established for none of 445 pairs (Figure 3).

Adding assumptions. The minimal-movement benchmark (column B) establishes no direction after 2012. Adding that out-of-zone voters resemble their station within ten points (column C) yields positive slopes in every election except 2017, but only because the benchmark confines most early voters to their own neighborhood. With mobility-calibrated movement instead (column D), the relation is again undetermined in 2022, 2024 and 2025 and narrowly positive in 2026. Adding the restriction on residents' early-vote composition (column E) raises the 2026 lower bound to 0.054 and yields a fragile positive slope in 2024, lost within constituencies or with a price-dependent benchmark (column F); 2022 and 2025 remain undetermined. Section 4.5 asks which relaxations these conclusions survive.

Gradient and contrast. The contrast concerns a different summary of the association. Under mobility-calibrated movement, only in 2026, and only with the third assumption, are the higher-price halves of districts established to vote more conservatively than the lower-price halves: by between 0.8 and 7.2 points, around a conventional estimate of 3.8 (Figure 2C). Within districts, then, the evidence for a price gradient is stronger than the evidence that higher-price and lower-price neighborhoods differ as groups.

A published allocation. Re-implementing Lee's (2026) allocation of pooled ballots reproduces its reported correlations and slope, which lies inside our main range. Its allocation nevertheless exceeds the residents available for pooled ballots in 47 of 155 constituencies, by a median of 16 voters (Online Appendix D). More fundamentally, one allocation cannot reveal sensitivity to the rule: it would return a single slope even in 2022 or 2025, where feasible allocations give both signs.

4.3 Beyond Gangnam, Seocho and Songpa

To assess whether the 2026 within-district result depends on Gangnam, Seocho and Songpa, we exclude their neighborhoods from the estimands. We recompute the regression and contrast weights for the 357 neighborhoods with price data among the 360 in the remaining 22 districts while retaining all accounting constraints (Table 3 and the orange bars in Figure 2).

Table 3. Associations with and without Gangnam, Seocho and Songpa

Election	Across districts: all	Across districts: without the three	Within districts, slope: all	Within districts, slope: without the three	Higher-minus lower-price half (points): all	Higher-minus lower-price half (points): without the three
A. Across districts: point values; within districts: ranges						
2012 (A)	0.137	0.057	[0.008, 0.157]	**[−0.021, 0.129]**	**[−1.7, 5.3]**	**[−2.6, 4.1]**
2017 (C)	0.060	0.025	**[−0.060, 0.085]**	**[−0.074, 0.071]**	**[−2.7, 2.9]**	**[−3.1, 2.3]**
2022	0.197	0.131	**[−0.026, 0.223]**	**[−0.042, 0.204]**	**[−3.2, 11.6]**	**[−3.9, 10.5]**
2024	0.172	0.097	[0.001, 0.206]	**[−0.027, 0.184]**	**[−1.6, 9.4]**	**[−2.6, 8.3]**
2025	0.137	0.080	**[−0.037, 0.194]**	**[−0.064, 0.173]**	**[−4.2, 9.4]**	**[−5.1, 8.4]**
2026	0.142	0.086	[0.054, 0.156]	[0.029, 0.139]	[0.8, 7.2]	**[−0.3, 6.2]**
B. 2026 sensitivity thresholds s (not slopes): largest additional departure at which the lower end stays positive; main setting s = 0.02*						
Mobility-calibrated, ±10	–	–	∞	0.103	0.049	0.010
Mobility-calibrated, ±20	–	–	0.087	0.001	none	none
Unrestricted movement, ±10	–	–	0.122	0.067	0.019	none
Unrestricted movement, ±20	–	–	0.006	none	none	none

Notes. Bold ranges include zero; their sign is not identified. Across districts: slope of the valid-vote share on log price across districts (point values, no allocation assumption). Within districts: ranges in z under the main assumptions (Table 2, column E); 2012 under column A and 2017 under column C, as those elections lack mobility data. Without the three: Gangnam, Seocho and Songpa dropped from the regression and the contrast, all accounting constraints kept. Panel B: s* with movement and the out-of-zone tolerance (±10 or ±20 points) held fixed; a relation is established at the main setting when s* exceeds 0.02. ∞: positive however much residents' early ballots depart; none: not positive for any s (Online Appendix Table C19).

Among election-day voters the slope outside the three districts is positive in 2022–2026 (Online Appendix Table C20). Among all residents, under the main assumptions, the 2026 within-district slope remains positive after the exclusion: its range is 0.029 to 0.139, compared with 0.054 to 0.156 for all of Seoul. In magnitude, a neighborhood whose apartment prices are twice those of another in the same district has a conservative roll-based share 2.0 to 9.6 points higher in the fitted relation; this describes the regression slope, not the effect of a price change or a difference found between every pair of neighborhoods. The median-split contrast ranges from −0.3 to 6.2 percentage points after the exclusion and the top-minus-bottom-tercile contrast from 0.1 to 7.9 (Online Appendix Table C23), so the group comparisons are borderline. Valid-vote shares yield the same conclusions about signs (Online Appendix Table C18). Excluding the three districts also removes the positive lower bound for the 2024 slope under the main assumptions and for the 2012 slope based on returns and rolls alone (Table 3).

The positive slope outside the three districts requires the restriction on residents' early-vote composition: under mobility-calibrated movement and the ten-point tolerance alone (column D), the slope without them ranges from −0.020 to 0.188. Holding movement and the tolerance fixed, it stays positive as long as residents' early ballots depart from their benchmark by no more than about 0.103 beyond their station's departure, about five times the main setting's 0.02; with unrestricted movement the threshold is 0.067. The result is more sensitive to the out-of-zone tolerance: with a 20-percentage-point tolerance and mobility-calibrated movement, the slope remains positive only up to an additional tolerance of 0.001,

below the main setting (Table 3, Panel B; Online Appendix Table C19). The conclusion beyond Gangnam is therefore less robust than the conclusion for all of Seoul, which survives a 20-point tolerance up to $s = 0.087$.

Under the main assumptions, the 2026 positive slope extends beyond Gangnam, Seocho and Songpa. It is not driven by any single remaining district: omitting each in turn keeps the lower bound positive. High-price neighborhoods nevertheless matter. Excluding the top 5% or 10% by price among the remaining neighborhoods lowers the bound to 0.0004 and -0.0093, respectively (Online Appendix Table C29). The result therefore extends beyond the three-district enclave while remaining sensitive to the upper end of the price distribution. It is compatible with economic interests, sorting and place identity operating elsewhere in Seoul (Bae and Joo 2020).

4.4 Information supplied by early-voting zones

The elections differ sharply in what they reveal, and one reason is administrative. In presidential elections the early-voting zone was the whole district; in 2026 it was the district-council constituency, about three neighborhoods. Because the 2026 returns can be analyzed with and without the zone rule, the role of the rule can be isolated in the same data (Table 4).

Table 4. Within-district slopes with and without statutory zones, 2026

Assumptions	Zones ignored: in-zone early ballots may be cast anywhere in the district	Statutory zones: only within the voter's council constituency
Returns and rolls only	**[−0.378, 0.519]**	**[−0.279, 0.439]**
Minimal movement, out-of-zone within 10 points	[0.072, 0.139]	[0.072, 0.138]
Uniform movement ($\eta = 0.1$), out-of-zone within 10 points	**[−0.039, 0.239]**	[0.014, 0.195]
Mobility-calibrated movement, out-of-zone within 10 points (D)	**[−0.191, 0.373]**	[0.010, 0.201]
Main assumptions (E)	[0.034, 0.174]	[0.054, 0.156]

Notes. Bold ranges include zero; their sign is not identified. Roll-based share z. The zones are the 155 district-council constituencies, about three neighborhoods each. Rows with uniform movement hold the amount of movement fixed, so the comparison isolates the zone rule; the mobility-calibrated rows recalibrate movement to the zone definition. Adding the council contest's published headcounts changes the statutory-zone ranges by at most 0.003 (Online Appendix Table C2).

With the uniform movement cap held fixed, allowing early ballots to move throughout the district yields a slope range of $[-0.039, 0.239]$. Restricting them to their statutory zones raises the lower bound above zero, to $[0.014, 0.195]$. The mobility-calibrated comparison gives the same change in sign status ($[-0.191, 0.373]$ versus $[0.010, 0.201]$), although its movement caps are recalibrated to each zone definition. Under the main assumptions, both ranges are positive and the zones narrow the range by about a quarter.

The fixed-cap comparison isolates information in the zone boundaries: votes and pooled-ballot totals remain unchanged, and only the permitted destinations of in-zone ballots change. It does not simulate a presidential election, in which some voters classified as out-of-zone in 2026 would instead be in-zone. The populations subject to the out-of-zone restriction also differ across election types. Table 4 thus demonstrates the informational value of zones within one election; it does not decompose the political and administrative causes of the 2025–2026 difference.

4.5 Sensitivity and robustness

Figure 4 summarizes the three-parameter sensitivity analysis. Each cell reports a threshold s^*: for example, 0.087 means that the slope remains positive while the extra composition tolerance is below about 8.7 percentage points. The main setting uses two points. The figure maps how far the assumptions can be relaxed: for each amount of movement and each out-of-zone tolerance, the largest additional departure s^* at which the slope stays positive. For all of Seoul in 2026, with a 10-point tolerance the slope stays positive at the main setting for every amount of movement, including unrestricted movement ($s^* = 0.122$). With a 20-point tolerance, s^* is 0.087 under calibrated and 0.006 under unrestricted movement (Online Appendix Tables C4 and C19). The 2026 conclusion therefore depends mainly on the out-of-zone tolerance, not on the mobile-phone calibration; at the main s it holds up to $\delta = 0.26$ for all of Seoul and $\delta = 0.18$ outside the three districts (Online Appendix Table C25). In 2024 the region is a narrow strip ($s^* = 0.021$ at the main movement and tolerance); in 2022 and 2025 there is none.

Price-aligned departures. The unresolved signs can be settled by a different assumption that limits how residents' early-vote departures align with neighborhood prices, rather than limiting each neighborhood separately (Online Appendix A.1 and Table C22). For all of Seoul, departure slopes at the scale of observed station patterns would establish positive gradients in the four sampled elections from 2022. Those patterns concern station voters or district averages, however, and do not measure the unobserved residential departures being restricted. Outside the three districts the conclusion is less uniform. Appendix C reports the thresholds and comparisons, including a restriction on out-of-zone departures alone (Table C25). These exercises identify assumptions that would settle the open signs; they do not establish that those assumptions hold.

Table 5 characterizes reversals. The 2022 and 2025 signs are unresolved even when residents' early votes depart from their benchmark only as much as their station's ballots do. Restricting movers' composition does not settle them because pooled ballots can instead support a reversal (Online Appendix Tables C5–C7 and C25). In 2026 no reversal exists under calibrated movement and the ten-point out-of-zone tolerance, even without restricting residents' in-zone composition. At a 20-point tolerance it requires an additional departure of about 9 points. These allocations describe what the assumptions allow, not estimated voting behavior.

Compatibility with observed counts. We assess how the result changes when composition tolerances are set relative to the minimum required by each jurisdiction's counts. Holding calibrated movement and $s = 0.02$ fixed, the largest jurisdiction-specific minimum feasible out-of-zone band is 0.031, 0.046, 0.061 and 0.073 in 2022, 2024, 2025 and 2026. We therefore replace the constant band by each jurisdiction's minimum plus a common excess tolerance (Online Appendix Table C27). At an excess of ten points, the 2026 slope remains positive for all Seoul and outside the three districts, with lower bounds 0.040 and 0.012. These jurisdiction-specific bands assess proximity to refutation and are not generally nested with the constant-band main setting (Online Appendix C).

Alternative specifications.

An alternative 2025 price measure leaves the sign conclusions in columns A, C, D and E unchanged. The comparison uses a separately documented official transaction snapshot and address mapping, with results also restricted to the neighborhoods priced in both sources (Online Appendix B.3 and Table C28).

The 2026 conclusion under the main assumptions holds with regressions weighted by H_i, mean prices, without the 43 neighborhoods with fewer than 30 transactions, and without the 5% or 10% most expensive

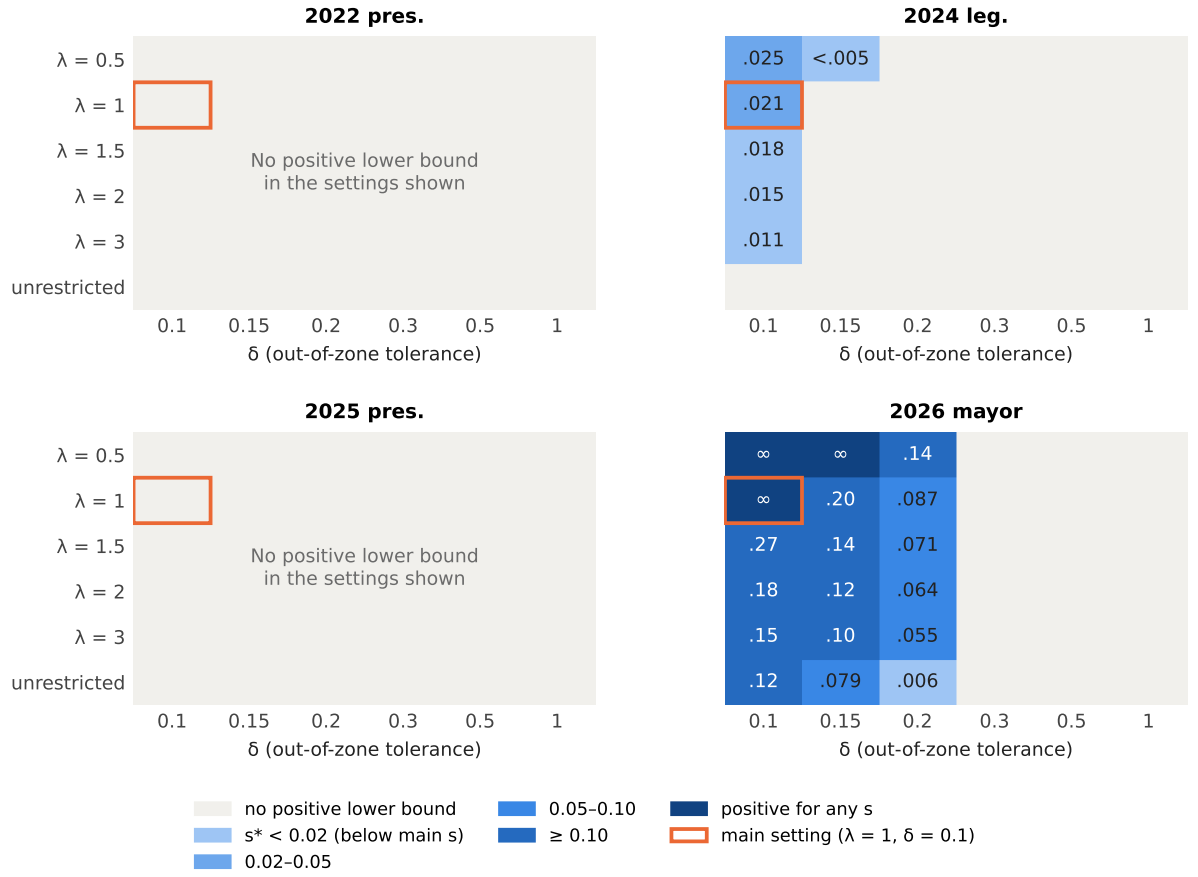


Figure 4: Where the within-district slope is positive. Each cell gives s^*, the largest additional departure of residents' early ballots from their neighborhood benchmark, beyond their station's own departure, at which the lower end of the range stays positive; amounts of movement are multiples λ of the mobility-calibrated scenario or unrestricted, and δ is the tolerance for out-of-zone voters. The main setting ($\lambda = 1$, $\delta = 0.1$, $s = 0.02$) is outlined: the slope is positive there if s^* exceeds 0.02. "∞": positive however much residents' early ballots depart; gray: no positive lower bound for any $s \geq 0$ among the settings examined. All of Seoul; numbers in Online Appendix Table C4, and selected settings without Gangnam, Seocho and Songpa in Table C19.

Table 5. Assumptions and allocations that permit a non-positive slope

Election	Out-of-zone voters within (points)	Largest additional tolerance s with a positive slope	First reversing allocation: movers minus stayers, conservative share (points)	Fewest residents' early ballots that must depart to reverse
2022	10	none	12 to 21	16,685
2022	20	none	14 to 18	15,340
2024	10	0.021	11 to 17	32,177
2024	20	none	9 to 14	30,103
2025	10	none	20 to 24	13,539
2025	20	none	21 to 23	12,512
2026	10	∞	no reversal	no reversal
2026	20	0.087	−2 to 24	69,384

Notes. Mobility-calibrated movement. s: residents' early ballots may depart from their neighborhood benchmark (election-day share shifted by the district's early-vote gap) by their station's departure plus s; ∞: positive however much they depart; none: not positive even when they depart only as much as their station. First reversing allocation: the one with the lowest slope at s just above that value; movers voted at another station of their zone, stayers at their own; range over price terciles. Last column: among allocations with a non-positive slope, the fewest residents' conservative early ballots departing from the benchmark (out-of-zone departures and concentration in Online Appendix Table C6). Each reversal is minimal only in its own metric, with other assumptions fixed.

neighborhoods, but it depends on the most expensive neighborhoods: without the top 10% the lower bound falls from 0.054 to 0.008, and with $s = 0.05$ it turns negative (Online Appendix Table C10). Resampling transactions moves the bounds by a few thousandths (Table C13).

A benchmark that varies with price within districts, motivated by larger early-vote gaps in higher-price districts in 2024–2026, changes the 2026 result only slightly (Table 2, column F), as does shifting the out-of-zone benchmark by the observed out-of-zone gap (Table C9). Valid-vote shares y lead to the same conclusions, except that column C does not establish the 2025 slope for them (Table C15), and the council contest's out-of-zone headcounts change the 2026 ranges by at most 0.003 (Table C2). Adjusting for neighborhood age composition measured in 2026 leaves the election-day slopes almost unchanged and the 2026 range among all residents positive ($[0.046, 0.160]$; $[0.024, 0.148]$ outside the three districts), but removes the fragile 2024 lower bound (Table C23).

5. Reporting reforms

The uncertainty is partly a product of what is published, and it could be reduced without publishing any individual vote. Out-of-zone early ballots already arrive at district commissions in individually tracked envelopes. *Residence reporting* would sort them by the voter's registered neighborhood before opening and publish aggregate counts by neighborhood. At-home and overseas ballots, which also arrive by mail, could in principle be reported the same way. *Headcount reporting* would publish how many residents of each neighborhood voted at each early station, with no vote choice.

Because the true allocation is unknown, we evaluate each change by the uncertainty it is guaranteed to remove: the largest range that could remain whatever the true allocation, computed as the largest difference in the within-district slope between two allocations that the new report could not tell apart. Table 6 reports, under the main assumptions, the current range and the largest range that could remain after each report.

Table 6. Guaranteed reductions in uncertainty under residence reporting

Election	Current range width	Proposal: out-of-zone early ballots by neighborhood, share removed	Proposal: maximum remaining width	All pooled modes by neighborhood, share removed	All pooled modes: maximum remaining width
2022	0.25	27%	0.18	57%	0.11
2024	0.20	30%	0.14	49%	0.11
2025	0.23	32%	0.16	61%	0.09
2026	0.10	57%	0.04	69%	0.03

Notes. All columns describe the width of the within-district slope range or the guaranteed fraction of that width removed. Main assumptions: Table 2, column E. Proposal: out-of-zone early ballots reported as aggregate counts by the voter's registered neighborhood, at-home and overseas ballots kept pooled. All pooled modes: out-of-zone early, at-home and overseas ballots by neighborhood. Maximum remaining width: the largest difference in the within-district slope between two allocations that the report could not tell apart, which bounds the range left whatever the true allocation. Counts are released exactly, without cell suppression. Suppression sensitivity: Online Appendix Table C26. Other assumptions and headcount reporting: Table C12.

Residence reporting of out-of-zone early ballots, with at-home and overseas ballots kept pooled, is the version we evaluate most closely. With counts released exactly, it is guaranteed to remove 27%, 30%, 32% and 57% of the current range in 2022, 2024, 2025 and 2026; reporting all pooled modes would remove 49–69% (Online Appendix Table C12). Neither version guarantees that a currently undetermined direction is revealed: in 2022, two allocations with slopes of -0.026 and $+0.026$ produce the same exact residence report (Table C16). Headcount reporting alone reveals little about how movers voted, but combined

with reporting all pooled modes it removes 63–72% of the range that remains without any movement assumption (Table C12).

Exact disclosure and a release with small cells suppressed supply different information. As a sensitivity exercise, suppose each out-of-zone neighborhood-by-candidate count below 20 is marked suppressed and every other count is released exactly. A conservative calculation bounds the remaining integer uncertainty and guarantees reductions relative to the current continuous range of at least 23.7%, 26.8%, 29.6% and 49.0% in the four elections (Online Appendix Table C26). The calculation relaxes equality of reports to a necessary condition for two integer allocations to generate the same suppressed report; its bounds need not be sharp. It does not cover complementary suppression of larger counts or certify the privacy of such a release. Those choices require a separate disclosure assessment, and could reduce the information gained further.

The guaranteed reductions characterize the information supplied by each reporting rule under the stated restrictions. Implementation would require sorting out-of-zone envelopes into neighborhood groups before opening and assessing which aggregate counts could be released safely. A supplementary release after the official count could preserve the timing of official results, but ballot secrecy and operational costs require evaluation. The counts would permit direct neighborhood comparisons of residents' early and election-day votes, without explaining why those groups differ. Headcount reporting measures movement without adding candidate totals. Each proposal should be judged by both the uncertainty it removes and the disclosure constraints it must meet.

6. Discussion and conclusion

Higher-price neighborhoods in Seoul vote more conservatively among election-day voters in 2012 and the four sampled elections from 2022, including outside Gangnam, Seocho and Songpa. The extension across voting modes is established in 2026 under the main assumptions, is sensitive to specification in 2024, and remains unresolved in 2022 and 2025. Even in 2026, the result outside the three districts depends on high-price neighborhoods, and the median-split contrast includes zero. These distinctions matter for explanations of housing politics: the observed gradient extends beyond the southeastern enclave, but evidence for a general separation between higher- and lower-price groups is weaker.

The comparison across voting modes changes what can be inferred from a neighborhood map. Election-day voters supply a directly observed residential pattern, but early voters are politically different and partly missing from residential returns. A conventional allocation supplies one answer to that missing-data problem. Our bounds show whether other allocations consistent with the same counts can reverse it, and how much restrictions on movement and composition narrow the answer. The finding of an unresolved sign is consequently informative about the evidence needed to establish the relationship; it does not imply that housing and voting are unrelated.

The analysis also identifies information that election administration could supply. Finer statutory zones can establish a sign that remains unresolved under the same movement cap and coarser boundaries. Exact residence reporting of out-of-zone ballots would guarantee reductions of 27–57% in the remaining range under the main assumptions, although it need not settle an unresolved sign. The suppression exercise shows that some of this gain can survive withholding small counts. Operational costs and disclosure risks still require assessment before implementing such a release.

For studies using published neighborhood returns, the practical implication is to state how unattributed ballots are handled and show whether the substantive conclusion depends on that choice. This applies to the price-level and price-change questions studied by Kim and Kang (2022) and Yun and Jun (2024), although we do not directly test their estimates. Explicit allocations such as Lee's (2026) make their rules visible; allocation bounds add a measure of the conclusions those rules leave open. Reporting the election-day relationship alongside the resident-based range gives readers an observed benchmark and a clear account of the extrapolation across voters.

Other electoral systems face different versions of this problem. German precinct returns omit postal ballots, Australian booth returns omit declaration votes, and many American precinct datasets must allocate absentee ballots reported at county level (Online Appendix B.5; Amos et al. 2017; Baltz et al. 2022). Taiwan ties voting to household registration and reports residence-based station returns. Japan combines ballots at the counting district, usually a municipality or ward, with no neighborhood results. Korea publishes finer rows but mixes residential origins within them. Applying the approach elsewhere requires small-area returns, candidate totals for pooled ballots and compatible information on residents' use of voting modes. These reporting differences determine which geographic questions the data can answer.

Earlier local elections would test whether the 2026 pattern recurs across candidates and turnout conditions. Historical age profiles and direct observations of early-voting movements could also sharpen the sensitivity analysis. For the six contests studied here, the contribution is specific: the paper establishes the geographic reach of the election-day gradient, measures the conditions for its extension across voting modes, and identifies reporting changes that would make that extension less dependent on assumptions.

Data availability

An anonymized replication package provides the analysis inputs, available source snapshots, code and commands for regenerating the tables and figures and checking the headline calculations; it will be deposited publicly upon acceptance. For 2025 prices, it includes the archived summary and a separately documented reconstruction from a new transaction snapshot. The original summary-generation materials are unavailable; Online Appendix B.3 documents this distinction and Table C28 reports the sensitivity comparison.

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